Patient AI Health Navigator

KANINI | Vertical Use Case Portfolio 2026

Use Case HC-03 | Healthcare — Provider Patient Engagement | Tier 3 Transformational Bet

Field	Value
Document Type	Vertical Use Case — Business Challenge & Market Assessment
Domain / Sub-domain	Healthcare / Provider Patient Engagement (between-visit navigation, transitions of care, chronic-condition self-management)
Target Geography	United States
Primary Economic Buyer	Health system Chief Digital / Chief Experience Officer, VP Population Health or Care Management, CMO/CNO for transitions-of-care programs; for payers, VP Care Management / Stars; for ACOs and large physician groups, the value-based care executive
Primary Users	Patients and caregivers (daily); care managers, transitional-care nurses, pharmacists and contact-center staff who work the escalation queue
Industry Spend Pool	Medication non-adherence associated with ~$100B/yr of avoidable US health care services [1] (up to $528.4B/yr for all non-optimized medication use [2]); 3.8 million adult 30-day readmissions in 2018 at an average $15,200 each [6]
Addressable Market	US patient engagement solutions: $7.4B [47] to $12.1B [48] (2025); AI-conversational slice estimated at ~$0.4B [49] to ~$3.7B (KANINI derivation from [48], see 3.1)
Existing KANINI Material	BC05_Patient_Health_Navigator_v_1.0 (KodeGenAI 2026 hackathon challenge, Use Case #19) — problem framing reused; all figures re-verified and several corrected (see 1.4)
Demo Data Basis	Synthea synthetic patients (FHIR R4 + CSV, clinical notes enabled) grounded with MedlinePlus Connect, openFDA/DailyMed labels and RxNorm; benchmarked against CMS readmission, NHAMCS ED and HCUP data (Appendix A)
Research Confidence	Medium — regulatory, payment and safety evidence is primary-sourced and current; market sizes rest on commercial analyst estimates that disagree by ~60%; most competitor outcome claims are vendor-reported
Version	1.0 — September 2026

Executive Summary

Between clinic visits, US patients are largely on their own: 30–50% of chronic-disease medications are not taken as prescribed [1], roughly one in five to one in four scheduled appointments is missed [12], and Medicare-payer stays carry a 30-day readmission rate of about 17 per 100 [7]. Hospitals still face Hospital Readmissions Reduction Program (HRRP) penalties of up to 3% of Medicare base payments [8], and from FY2027 Medicare Advantage patients count toward those measures [11]. New money now funds between-visit work: Advanced Primary Care Management (APCM) codes from 2025 [20], cheaper remote-monitoring codes in 2026 [23], and the CMS ACCESS model, which began paying outcome-aligned fees to technology-enabled care organizations on July 5, 2026 [24]. Generative AI can now hold grounded conversations, but 7.1% of unedited LLM drafts in one study risked severe harm [38]. The market is crowded. Epic now ships a patient AI assistant inside MyChart [54], and well-funded AI-native vendors hold most large health systems [51][52]. What remains open is a narrower, billing-aligned and auditable navigator for systems that don't run Epic, for value-based groups, and for payers.

1. The Business Problem

1.1 Problem Statement

US health care expects patients to run their own care between visits. They have to understand a new diagnosis, take four to ten medications on schedule, notice warning signs, keep follow-up appointments, and judge whether a symptom at 10 p.m. means the emergency department (ED), an urgent-care visit, or a call to the clinic in the morning. The system gives them little structured help to do this. Care is delivered in visits, and the time between visits, which is where most chronic-disease outcomes are decided, is mostly unmanaged.

The costs show up in three places:

- **Medication non-adherence.** Among adults with chronic illness, 30–50% of medications are not taken as prescribed. US estimates link non-adherence with about 125,000 deaths, 10% of hospitalizations and roughly $100B of health care services each year [1]. The often-quoted "$100–289B" range comes from studies from the 2000s that were summarized in a 2012 AHRQ-funded review [3][4], so it is dated. A broader 2018 model of all *non-optimized* medication use, which includes non-adherence, puts the cost at $528.4B (range $495.3–672.7B) in 2016 dollars [2]. A systematic review of 79 studies found disease-specific costs of non-adherence of $949 to $44,190 per patient per year [5].
- **Readmissions and poor transitions.** In 2018 there were 3.8 million adult 30-day all-cause readmissions, at a 14% rate and an average cost of $15,200 each [6]. The Medicare 30-day readmission rate was 17.0 per 100 index stays in 2020, the highest of any payer [7]. HRRP measures six conditions and procedures: AMI, COPD, heart failure, pneumonia, CABG, and elective hip/knee replacement. It cuts Medicare base payments by up to 3% [8].
- **Missed appointments and misdirected acute care.** A systematic review of 105 studies put the average no-show rate at about 23% [12]. Primary care sits at the low end and behavioural health at the high end. In an August 2025 MGMA poll, 27% of practices reported that no-shows were rising [13]. The best-known US estimate says 13.7–27.1% of ED visits could be handled at urgent-care centres or retail clinics, a saving of about $4.4B [14]. That study is from 2010, and estimates like this depend heavily on how "avoidable" is defined. In 2022 there were 155.4 million US ED visits [15].

A concrete case: a 72-year-old man goes home after a stay for acute on chronic systolic heart failure (ICD-10 **I50.23**). He takes new furosemide 40 mg tablets (RxNorm **313988**), a changed carvedilol dose (RxNorm **200031**), and apixaban (RxNorm **1364445**), plus his existing metformin 500 mg (RxNorm **861007**) for type 2 diabetes. His paperwork runs to several pages and asks him to weigh himself every day (LOINC **29463-7**), limit salt and fluids, and see his primary care physician within 7–14 days. By day 3 he has stopped the carvedilol because he feels dizzy. He doesn't know whether the 1 kg of weight he has gained matters, and nobody has booked his follow-up visit. Under Transitional Care Management (TCM), someone from the practice has to contact him within 2 business days of discharge. The face-to-face visit must happen within 7 days for high-complexity TCM (CPT **99496**) or 14 days for moderate complexity (CPT **99495**) [19]. Most practices struggle to do this reliably at scale. A navigator that knows *his* medication list, *his* discharge instructions and *his* appointment status could close these gaps. A generic chatbot or a static PDF cannot.

1.2 Why This Problem Persists

- **Care is episodic and paid by the visit.** Fee-for-service pays for visits, not for the days between them. Care-management codes exist, but they carry time thresholds, consent rules and documentation requirements. CMS itself said the APCM codes were built without time thresholds "to reduce the administrative burden" of the older codes [20]. Many practices never set up the between-visit infrastructure these codes require, such as 24/7 access, a designated care-team member and electronic care plans [21].
- **Patient materials are generic and hard to read.** In the last national health-literacy assessment (2003, now dated), 36% of adults had Basic (22%) or Below Basic (14%) health literacy [16]. General literacy is getting worse. In 2023, 28% of US adults scored at or below Level 1 in literacy, up from 19% in 2017 [17]. Discharge instructions and medication guides are still written far above that level and are not tailored to the patient's own regimen.

- **Portals give access to data, not guidance.** Portal adoption is no longer the bottleneck. In 2024, 77% of individuals were offered online access to their records and 65% used it at least once [18]. Most portal use is to view results and notes. The bigger problem is fragmentation: 59% of individuals had more than one portal, and only 7% used an app to combine them [18]. BC05's "15–30% active portal utilization" figure is out of date.
- **Staff can't scale to the need.** Nurse lines, care managers and call centres are expensive and hard to staff. The AAMC projects a shortage of up to 86,000 physicians by 2036 as the 65+ population grows 34.1% [28]. Adding a patient-message channel without triage just moves work into clinicians' inboxes. In a randomized quality-improvement study, AI-drafted replies *increased* physician read time by 21.8% and did not reduce reply time [41].
- **Safety and liability make providers cautious.** Unsupervised AI that talks to patients is risky. Physicians judged that 7.1% of unedited GPT-4 drafts to cancer-patient messages risked severe harm and 0.6% risked death. Most of the harmful drafts got the *acuity* of the situation wrong [38]. A 1,298-person randomized study found that members of the public using LLMs identified the relevant condition in fewer than 34.5% of scenarios, no better than controls, even though the models alone got 94.9% right [39]. ECRI ranked misuse of AI chatbots as the #1 health technology hazard for 2026 [37].
- **Evidence of benefit is thin, and effect sizes are shrinking.** A meta-analysis of 42 randomized trials found that transition interventions reduced 30-day readmissions (pooled RR 0.82). Complex interventions that build the patient's capacity for self-care worked best, and more recent trials showed smaller effects than older ones [42]. Simple reminder bots are unlikely to repeat those results. Adherence research likewise favours personalized, recurrent outreach and refill-prompt calls, which have shown absolute adherence gains of up to 33% [1].

1.3 Rationale and Industry Context

- **Readmission penalties are going up and now cover Medicare Advantage patients.** In FY2026, 240 hospitals (8.1%) face HRRP penalties of 1% or more, up from 208 (7%) in FY2025. It is the first increase in five years [9]. From FY2027, all six readmission measures include Medicare Advantage beneficiaries, and the performance window shrinks to two years (July 1, 2023 – June 30, 2025 for FY2027) [11]. Improvements in transitional care will therefore show up in penalties sooner.
- **New codes pay for between-visit work.** The APCM codes (G0556/G0557/G0558) began on January 1, 2025. National rates are $15.20, $48.84 and $107.07 per patient per month. The codes require 24/7 access, continuity, digital communication channels and timely follow-up after transitions [20][21]. For CY2026, CMS added behavioural-health add-ons for APCM [22], and new remote-monitoring codes pay for 2–15 days of data (99445, ~$47) and 10-minute management (99470, ~$26) (vendor-compiled rates) [23]. APCM can't be billed in the same month as CCM, PCM or TCM for the same patient [21], so navigation workflows have to be designed around billing rules.
- **CMS now pays for technology-enabled chronic care.** The CMS Innovation Center's ACCESS model is a 10-year voluntary model that began on July 5, 2026. It pays recurring outcome-aligned payments to Medicare Part B-enrolled organizations managing cardio-kidney-metabolic, musculoskeletal and behavioural-health conditions. Heart failure and COPD tracks start on April 1, 2027. Referring primary care physicians can bill a new co-management payment [24][25]. More than 160 organizations are participating [24], most of them digital health companies without prior Medicare experience [26]. This affects the use case in two ways: it creates a funded buyer and a new group of competitors.
- **Payers have Star Ratings reasons to care about adherence.** Three PQA medication-adherence measures (diabetes, RAS antagonists, statins) are in the 2027 Part D Star Ratings [27]. That gives Medicare Advantage plans a direct reason to fund between-visit adherence navigation.
- **Patients are already asking chatbots.** About 17% of adults used AI chatbots at least monthly for health information in mid-2024. Only 29% trusted them to provide reliable health information [44]. OpenAI launched ChatGPT Health on January 7, 2026, with medical-record connectivity through b.well [77]. Patients are going to use AI for health questions either way. A provider-governed navigator grounded in the patient's own record is the safer alternative.
- **Rules for AI talking to patients are now in force.** California AB 3030 (signed September 28, 2024) requires a generative-AI disclaimer and instructions for reaching a human on AI-generated clinical communications, unless a licensed clinician reviews them first [29]. Texas TRAIGA (effective January 1, 2026) requires providers to disclose AI use in health care

services or treatment [30]. HHS Section 1557 requires covered entities to identify and mitigate discrimination risk in patient care decision support tools [33]. FDA updated its Clinical Decision Support and General Wellness guidance in January 2026 [31][32]. Federal policy is also pushing back on state AI laws: Executive Order 14365 (December 11, 2025) set up a federal AI Litigation Task Force to challenge "onerous" state AI laws [36]. The rules are moving, so a compliance-configurable design is worth more.

1.4 Scale of the Problem (Key Metrics)

Metric	Value	Source
Share of chronic-disease medications not taken as prescribed	30–50%	[1]
Annual US cost associated with medication non-adherence	~$100B (services); widely quoted $100–289B range is based on pre-2010 studies (dated)	[1][3] [4]
Annual cost of all non-optimized prescription drug use (incl. non-adherence)	$528.4B (range $495.3–672.7B), 2016 US$	[2]
Disease-specific annual cost of non-adherence per patient	$949–$44,190 (2015 US$)	[5]
Adult 30-day all-cause readmissions (2018)	3.8 million; 14% rate; $15,200 average cost	[6]
Medicare-payer 30-day readmission rate (2020)	17.0 per 100 index admissions (all payers 13.9)	[7]
HRRP maximum penalty	3% of Medicare base operating DRG payments	[8]
Hospitals with HRRP penalty ≥1% (FY2026 vs FY2025)	240 (8.1%) vs 208 (7.0%)	[9]
Most recent verified total annual HRRP penalty	~$320M across 2,273 hospitals (FY2023)	[10]
Average appointment no-show rate	~23% (systematic review, 105 studies; dated 2018)	[12]
Practices reporting rising no-shows (2025)	27% (MGMA poll, n=265)	[13]
ED visits treatable in urgent care / retail clinics	13.7–27.1% (2010 estimate, dated; definitions vary widely)	[14]
US ED visits (2022)	155.4 million (47.3 per 100 persons)	[15]
Adults with Basic or Below-Basic health literacy	36% (2003 NAAL, dated)	[16]
Adults at or below Level 1 literacy (PIAAC 2023)	28% (up from 19% in 2017)	[17]
Individuals offered / using online record access (2024)	77% offered; 65% accessed at least once	[18]
Individuals with more than one portal / using an aggregator	59% / 7%	[18]
Unedited LLM patient-message drafts judged to risk severe harm	7.1% (death: 0.6%), n=156	[38]
Adults using AI chatbots monthly for health info / trusting them	17% / 29% (2024)	[44]

Re-verification of BC05 figures. BC05's "$150B avoidable utilization" headline adds together estimates built on different methods, and this review could not trace it to one source, so it has been dropped. The "$150B cost of missed appointments" figure shows up only in secondary and vendor material and has no primary study behind it. It is not used here and is listed in 5.4. The "18–23% no-show" range has been replaced by the systematic-review average of ~23%, with specialty variation noted [12]. The "13–27% avoidable ED visits" figure is confirmed as 13.7–27.1% but comes from a 2010 study [14]. The "15–30% portal utilization" figure has been corrected to 65% [18]. The "$2–5M nurse-line cost" figure could not be verified and has been removed.

2. High-Level Objective

2.1 Solution Vision

The Patient AI Health Navigator is a between-visit assistant that a health system, physician group or health plan runs under its own brand, over SMS, web chat, portal and (in production) voice. It knows the patient's own record: active conditions, current medications, discharge instructions, care plan and upcoming appointments. It uses that record to explain, remind, check in and route the patient to the right next step. It reaches out after discharge and when refills lapse, and it answers the patient's questions at any hour in plain language at a sixth-to-eighth-grade reading level.

The navigator does not diagnose, prescribe, change doses or give an opinion on lab values. **It keeps a human in the loop by design.** Deterministic rules, not the language model, decide when a conversation must go to a human. Anything above a defined acuity level goes to a licensed care-team member or triggers an emergency instruction. Clinicians review escalated conversations, and every conversation turn is logged for audit. The disclosure language required by AB 3030 and Texas TRAIGA is built in [29][30]. For the business, the navigator turns transitional-care and care-management obligations into completed, documented, billable work. It scales the care team's reach without adding to clinician inboxes.

2.2 Solution Approach: Record-Grounded Conversational Agent with Deterministic Clinical Guardrails and Care-Team Escalation

The approach puts deterministic safety rules and structured data first, with AI assisting. The LLM writes plain-language explanations and runs the conversation. Structured rules decide what the navigator may say, when it has to escalate, and which billing-relevant events it records. This design responds directly to the evidence that LLM failures are mostly failures to judge acuity [38] and failures in how people interact with the model [39].

Layer	What It Does	Examples
Patient Context	Assembles a per-patient profile from FHIR R4 resources (Patient, Condition, MedicationRequest, CarePlan, Appointment, Observation, DocumentReference for the discharge summary) and pharmacy-fill data, and passes only that profile to the conversation.	Patient asks "What are my water pills for?" The navigator answers from MedicationRequest furosemide 40 mg (RxNorm 313988), prescribed at discharge for I50.23, and not from a generic drug list.
Knowledge Grounding	Pulls condition and drug education from authoritative public sources keyed to the patient's codes and cites them. It never answers drug questions from model memory alone.	MedlinePlus Connect lookup on ICD-10 I50.23 returns the NLM "Heart Failure" topic; side-effect answers are grounded in the openFDA/DailyMed label for apixaban (RxNorm 1364445).
Proactive Outreach Engine	Schedules check-ins keyed to events: discharge (within 2 business days, supporting the TCM contact element [19]), day 7/14 follow-up visit confirmation, refill-gap detection from fill data (proportion-of-days-covered logic aligned to PQA adherence measures [27]), and pre-visit preparation.	Day 2 after discharge: "Have you been able to pick up your new medicines? Your follow-up with your primary care doctor isn't booked yet. Would you like me to find times?" A statin refill that is 10 days overdue triggers an adherence check-in and flags cost barriers.
Deterministic Clinical Guardrails	Rule engine that runs *before and after* the LLM. It matches red-flag symptoms, scope violations (diagnosis, dose change, lab interpretation) and self-harm language, and overrides the model's output. The rules are versioned and signed off by the organization's clinical governance group.	"Chest pain", "can't catch my breath", FAST stroke signs or suicidal ideation get a fixed emergency script (call 911 / 988) and a care-team alert. "Can I stop my carvedilol? I'm dizzy" gets no advice to stop; the message goes to the nurse queue as a same-day callback with the patient's words and medication list attached.
Human Escalation and Care-Team Console	Routes escalations to the right queue (nurse line, pharmacist, scheduler, social worker) with a structured summary. It records who responded and when, so the organization can prove clinician involvement.	Escalation card: "HF patient, day 3 post-discharge, reports dizziness and self-discontinued carvedilol 6.25 mg (RxNorm 200031); weight +1 kg since discharge per patient report; no follow-up booked."

Layer	What It Does	Examples
Governance and Evidence	Audit log; disclosure banners (AB 3030, TRAIGA) [29][30]; Section 1557 §92.210 bias review of any risk-scoring inputs [33]; offline evaluation harness (red-team scenarios, acuity-accuracy tests, reading-level checks); outcome dashboards (readmissions, TCM completion, adherence).	Monthly report: 100% of red-flag test prompts escalated; median Flesch-Kincaid grade 6.8; 30-day readmission rate for enrolled HF patients vs matched non-enrolled cohort.

2.3 How the System Would Work (Functional Flow)

The flow below follows a patient discharged after a heart-failure stay, then the transitional-care nurse who works the escalations.

1. At discharge, the hospital enrols the patient. The navigator loads the discharge summary, medication reconciliation, care plan and follow-up orders, and sends a text introduction that includes the required AI disclosure and a way to reach a human.
2. Within 24–48 hours, the navigator checks in. It confirms the patient picked up the medications, explains in plain language what each new drug is for, and asks whether the patient has symptoms from their own discharge instructions (daily weight, swelling, breathlessness).
3. The patient asks questions in their own words, such as "Why two blood-pressure pills?" or "Can I eat soup?" The navigator answers from the patient's record and grounded education content, and checks understanding ("Would it help if I explained that differently?").
4. If a message hits a guardrail (a red-flag symptom, a request to change a dose, or something outside scope), the rule engine replaces the model's answer with a fixed safe response. It then creates an escalation card for the right queue, marked emergency, same-day or routine.
5. The transitional-care nurse works the queue from the console. Each item has a summary and the full conversation. The nurse calls the patient, documents the action, and the contact is recorded against the TCM or APCM episode.
6. Over the next 30 days, the navigator confirms the follow-up visit (and books it through scheduling where integrated), sends a reminder the day before, runs refill-gap checks, and closes the episode with a short satisfaction and understanding check.
7. The population-health manager reviews the dashboard: enrolment, response rates, escalations by type, TCM contact compliance, follow-up visit completion, and 30-day readmissions against a comparison cohort.

2.4 What Success Looks Like

Success Metric	Target	Why It Matters
Red-flag escalation recall (test suite + live audit)	100% of scripted emergency scenarios escalate; zero missed live emergencies on audit	Getting acuity wrong is the dominant LLM harm mode [38]; a missed chest-pain escalation is a catastrophic failure
Record-grounding accuracy	100% of medication and appointment facts match the source record	A navigator that invents a medication is a safety event, not a usability issue
TCM 2-business-day contact completion	Baseline (organization-specific, often manual and incomplete) → ≥90% of eligible discharges	Required element for billing 99495/99496 [19]; turns navigation into documented revenue
7/14-day post-discharge follow-up visit completion	+15–20 percentage points vs baseline (hypothesis to validate)	Timely follow-up is a core transitional-care element and a TCM billing condition [19]
30-day readmission rate, enrolled HRRP-condition patients	10–15% relative reduction vs matched cohort (hypothesis; meta-analytic effect of transition interventions is RR 0.82 [42])	Directly affects HRRP exposure [8][9], and from FY2027 includes MA patients [11]

Success Metric	Target	Why It Matters
Adherence (PDC ≥80%) for statins / RASA / diabetes drugs in enrolled members	+5–10 percentage points vs baseline (hypothesis; refill-reminder interventions have shown up to 33% absolute gains [1])	Part D Star Ratings adherence measures [27]; the payer ROI case
Share of patient questions resolved without staff	≥60% resolved by navigator with no escalation, while escalation recall stays at 100%	Care-team capacity; vendor benchmarks are higher but vendor-reported [56]
Reading level of responses	Median Flesch-Kincaid grade 6–8	28% of adults are at or below Level 1 literacy [17]
Clinician inbox impact	No net increase in clinician message minutes per 1,000 enrolled patients	Avoid the pattern where AI drafts added read time [41]
Patient trust	≥80% of enrolled patients rate the navigator helpful; opt-out <10%	Only 29% of adults trust chatbots for health information [44]; trust has to be earned in each deployment

3. Market Analysis

3.1 Market Size and Growth

No analyst tracks "AI health navigation" as its own category. The nearest measured markets are patient engagement solutions (broad: portals, outreach, education, remote monitoring) and healthcare chatbots or AI agents (narrow). The estimates below disagree a lot, and each is a commercial analyst estimate built on its own definitions.

Estimate	Scope	Base Value	Forecast	CAGR	Source
Precedence Research	US patient engagement solutions	$7.40B (2025)	$27.83B (2035)	14.16%	[47]
Mordor Intelligence	US patient engagement solutions	$12.12B (2025); $13.36B (2026)	$22.75B (2031)	11.24%	[48]
Precedence Research	North America healthcare chatbots	$0.41B (2025)	$3.34B (2035)	23.33%	[49]
MarketsandMarkets	Global AI agents in healthcare	$1.11B (2025)	$6.92B (2030)	44.1%	[50]
Menlo Ventures (investor survey, n>700)	US health care spend on AI tools, all uses	$1.4B (2025); patient engagement growing ~20x YoY	—	—	[46]

Reading the spread. The two US patient-engagement estimates differ by about $4.7B, or roughly 64% of the lower figure. The difference mostly comes from whether remote monitoring, portals and services are counted. Mordor says "AI-driven engagement" is 30.51% of its market, and that providers are 61.23% of spend while payers are the fastest-growing segment (15.06% CAGR) [48]. Applying that share to the Mordor total gives a **KANINI-derived ceiling of ~$3.7B** for AI-driven engagement in 2025. The narrow chatbot estimate of $0.41B [49] is the floor. **KANINI working SAM: $0.4B–$3.7B (2025), growing 11–23% a year.** This range is wide on purpose, because we have no independent bottom-up survey of spend on this category.

Bottom-up cross-check (funding pools rather than software spend). Care-management codes set how much a provider can afford to spend. As an illustration only: a group with 2,000 attributed Medicare patients who have two or more chronic conditions, billing APCM G0557 at the 2025 national rate of $48.84 [21], would collect about $1.17M a year in gross fees (2,000 × $48.84 × 12, KANINI arithmetic). A navigator that makes the required access and communication elements feasible competes for part of that fee pool. On the hospital side, readmissions cost about $15,200 each [6], so avoiding even 100 readmissions a year is worth about $1.5M in avoided cost for a risk-bearing organization. For a fee-for-service hospital, the value is avoided HRRP penalty and freed capacity.

3.2 Buyer Segments and Buying Dynamics

Segment	Economic Buyer	Primary Motive	Typical Sales Cycle / Procurement	Build vs Buy Tendency
Large Epic-based academic and integrated systems	CDO / CMIO / VP Patient Access	Access, contact-centre deflection, portal engagement; readmissions secondary	9–18 months; IT security, AI governance committee, EHR vendor alignment	Increasingly "wait for Epic" (Ask Emmie, MyChart AI) [54] or pilot one AI-native vendor; some build on cloud LLM platforms
Mid-size and community hospitals (often Oracle Health, MEDITECH, or Epic via Community Connect)	CFO / CNO / VP Quality	HRRP exposure, nurse staffing, TCM completion	6–12 months; budget-constrained; wants ROI in <12 months	Buy; limited in-house AI engineering
Primary care groups, ACOs, MSOs, FQHCs	Value-based care executive / practice administrator	APCM/CCM/TCM revenue, shared-savings performance, ACCESS participation [24]	3–9 months; per-member-per-month pricing preferred	Buy or partner; very limited IT staff
Medicare Advantage and Medicaid managed-care plans	VP Care Management / Stars / Quality	Part D adherence Stars measures [27], readmission cost, member experience	6–12 months; RFP-driven; delegated vendor oversight	Buy (HealthEdge Wellframe and similar platforms are incumbents [61]); some build

Segment	Economic Buyer	Primary Motive	Typical Sales Cycle / Procurement	Build vs Buy Tendency
Specialty service lines (oncology, maternal, orthopaedics, cardiology)	Service-line leader	Episode outcomes, patient experience, bundled or episode payments	3–6 months departmental pilots	Buy point solutions (Memora-style programs [56])

How buying actually works. The economic buyer and the user are different people, and the user is a patient who has no budget. Buyers ask for three things: (1) proof of safety, meaning a clinical governance sign-off, red-team results and a clear escalation design; (2) EHR integration evidence, which in practice means Epic first, then Oracle Health and MEDITECH; and (3) ROI they can see within a fiscal year. Security review normally requires SOC 2 Type II and, more and more often, HITRUST. Competitors advertise both (e.g., Artera [57], Hippocratic AI [52]). Patient-facing AI also goes through the organization's AI governance committee, and California and Texas deployments need disclosure workflows [29][30]. Payers buy differently from providers. Payers can't read the provider EHR in real time, so they work from claims and pharmacy data. They measure success in Stars and medical loss ratio. They run formal RFPs with delegated-vendor oversight. Providers buy on workflow fit and EHR integration, and care about TCM/APCM revenue and HRRP penalties.

3.3 Market Drivers and Headwinds

Driver / Headwind	Type (Regulatory / Economic / Technology / Labour)	Impact	Source
HRRP penalties rising for the first time in five years; MA patients included from FY2027	Regulatory / Economic	Driver — bigger, faster financial link to transitional care	[9][11]
APCM codes (2025) and APCM behavioural-health add-ons (2026) fund continuous primary care	Regulatory / Economic	Driver — recurring monthly revenue for between-visit access	[20][21] [22]
New RPM codes for 2–15 days and 10-minute management (2026)	Regulatory / Economic	Driver — makes short post-discharge monitoring episodes billable	[23]
CMS ACCESS model launched July 2026 with 160+ participants; HF and COPD tracks from April 2027	Regulatory / Economic	Driver and competitive threat — funds outcomes, and draws in many new digital competitors	[24][26]
Part D Star Ratings adherence measures	Regulatory / Economic	Driver for payer segment	[27]
Clinician shortage (up to 86,000 physicians by 2036)	Labour	Driver — care teams need to reach more patients per clinician	[28]
LLM capability plus consumer use of chatbots (17% monthly) and ChatGPT Health launch	Technology	Driver — patients expect conversational help; providers want a governed alternative	[44][77]
Documented LLM harm modes (acuity errors; human–LLM interaction failures) and ECRI #1 hazard	Technology / Regulatory	Headwind — longer governance reviews; guardrail and evaluation costs	[37][38] [39]
State AI disclosure laws (CA AB 3030, TX TRAIGA) vs federal pre-emption push (EO 14365)	Regulatory	Headwind (uncertainty) — compliance must be configurable per state	[29][30] [36]
Section 1557 §92.210 obligations for patient care decision support tools	Regulatory	Headwind — bias review of any risk-scoring or triage inputs	[33]
FDA CDS and General Wellness guidance (Jan 2026): patient-facing software that informs clinical decisions can be a device	Regulatory	Headwind — keeps navigators on the "education/navigation, not triage-diagnosis" side	[31][32]
FTC scrutiny of consumer health data (Health Breach Notification Rule amendments effective July 29, 2024) and of AI chatbots (6(b) inquiry, September 2025)	Regulatory	Headwind — non-HIPAA consumer channels and engagement-driven chatbot design draw enforcement attention	[34][35]

Driver / Headwind	Type (Regulatory / Economic / Technology / Labour)	Impact	Source
Low public trust: 60% uncomfortable with provider reliance on AI; 29% trust chatbots for health info	Economic (demand)	Headwind — enrolment and opt-out risk	[43][44]
Epic embedding patient AI (Emmie) in MyChart	Economic / Technology	Headwind for third-party vendors in Epic accounts	[53][54]

3.4 Technology and Adoption Trends

- **Health systems are moving from pilots to purchases.** An investor survey of more than 700 health care executives found that 22% of organizations had implemented domain-specific AI tools in 2025, about seven times more than in 2024. Spending reached $1.4B, led by ambient documentation ($600M) and coding ($450M). Patient engagement is smaller but reportedly growing about 20x year over year [46]. (Investor-sourced; treat as directional.)
- **The EHR vendor is moving in.** Epic's patient assistant, Emmie, handles scheduling by text message, plain-language result and after-visit-summary explanations, and billing questions [53]. In March 2026, Epic said "Ask Emmie" had launched at Sutter Health to answer patient health questions in the context of the medical record, with broader availability planned. It also said more than 85% of its customers use Epic AI (vendor-reported) [54].
- **AI-native vendors are scaling voice agents.** Hippocratic AI reports more than 250 million patient interactions and use cases that include post-discharge readmission prevention and chronic care. It says it does not diagnose or prescribe (vendor-reported) [51]. It raised $141M at a $1.64B valuation in January 2025 and $126M at a $3.5B valuation in November 2025 [52].
- **Patient-communications platforms are becoming agent platforms.** Artera (2B+ interactions a year, 1,000+ organizations) and Luma Health (1,000+ organizations; "Navigator" conversational agent) have added agentic AI on top of existing messaging reach (vendor-reported) [57][58].
- **Published evidence is still early and mixed.** A 2025 systematic review of 23 studies found GenAI patient-message drafts empathetic and comparable in quality to human drafts. It also found inconsistent performance, safety risks and little real-world use [40]. In live deployment, AI drafts did not save physicians time [41]. Rigorous randomized evidence that conversational AI navigators reduce readmissions or improve adherence has not been published. Current claims are vendor case studies.
- **Patients' views are nuanced.** A KLAS–Luma survey of more than 1,000 US adults (July 2025) found patients most comfortable with AI for scheduling, check-in and billing, and less trusting of AI for diagnosis or treatment. Most wanted human supervision (vendor co-sponsored) [45]. Pew found 79% would not want an AI chatbot for mental health support [43].

4. Competitive Landscape

4.1 Competitor Categories

- **EHR-native incumbents:** Epic (MyChart + Emmie) [53][54]. It has the strongest distribution in its own accounts, and its patient AI is built into the portal patients already use.
- **AI-native patient-facing agent companies:** Hippocratic AI [51], K Health (PatientGPT intake plus virtual primary care for health systems) [69]. Well funded and quick to deploy voice and chat agents.
- **Care-navigation and digital care-program platforms:** Memora Health (now Commure) [55][56], Get Well (now SAI Group) [60], Amwell Automated Care Programs (formerly Conversa) [63][64], care.coach (avatar plus human health advocates) [65].
- **Patient-communication and access platforms adding agents:** Artera [57], Luma Health [58], Notable [70], Cedar (financial-only, Kora voice agent) [74].
- **Digital front door and virtual triage:** Clearstep (Schmitt-Thompson protocols) [66], Fabric (formerly Zipnosis and Gyant) [67], Infermedica [68], Ada Health [78].
- **Payer member engagement and care management:** HealthEdge Wellframe [61], Lirio (behavioural-science "precision nudging") [62], Laguna Health (conversational AI for payer care teams) [75].
- **Chronic-condition digital therapeutics and programs:** Welldoc (11 FDA 510(k) clearances) [71], Omada Health (public since June 2025) [72], Lightbeam (deviceless RPM via SMS/IVR/conversational AI, population health) [73].
- **Consumer general-purpose AI:** ChatGPT Health [77]. It isn't sold to providers, but it shapes what patients expect and competes for their attention.
- **In-house (DIY):** Large systems building on cloud LLM platforms and EHR APIs. How common this is could not be quantified.

4.2 Competitor Profiles

Company	Category	Target Segment	Core Offering for this Problem	Pricing Model (if public)	Strengths	Weaknesses / Complaints	Recent Moves (2024–2026)
Epic (MyChart / Emmie)	EHR-native incumbent	Epic health systems	Emmie: text-based scheduling, plain-language results and AVS, billing Q&A, care-plan reminders; "Ask Emmie" record-grounded Q&A [53] [54]	Not public	Native EHR data and workflow; installed patient base; reported Ochsner and Rush results (vendor-reported) [53]	Only for Epic customers; Ask Emmie was early (Sutter launch) as of March 2026 [54]; limited evidence of proactive post-discharge navigation or cross-EHR use	Ask Emmie launch; Epic says >85% of customers use Epic AI (Mar 2026) [54]
Hippocratic AI	AI-native agents	Health systems, payers, life sciences	Voice agents for post-discharge follow-up, chronic care, screening outreach, intake; explicitly no diagnosis or prescribing [51]	Not public	Scale (250M+ interactions); clinician-validated testing (775K test calls, 7,700+ clinicians); HITRUST e1 (vendor-reported) [51] [52]	Outcome claims (e.g., 30% readmission reduction, 12x ROI) are vendor case studies without independent peer review [51]; voice-first	$141M Series B (Jan 2025, $1.64B); $126M Series C (Nov 2025, $3.5B); acquired Grove AI (Jan 2026) [52]
Memora Health (Commure)	Care-navigation platform	Health systems, specialty service lines	Digitized care programs: transitions of care, CCM, oncology, maternal, surgical; automated check-ins and symptom management [56]	Not public	Clinical-program depth; named customers (Penn Medicine, Intermountain, Boston Medical Center) [56]; reports 85% of patient inquiries resolved by AI (vendor-reported) [56]	Now one part of Commure's broad portfolio (ambient AI, RCM, call centre), so the focus of the navigation roadmap may shift [55]	Acquired by Commure (Dec 20, 2024) [55]

Company	Category	Target Segment	Core Offering for this Problem	Pricing Model (if public)	Strengths	Weaknesses / Complaints	Recent Moves (2024–2026)
Artera	Patient communications + agents	Health systems, FQHCs, federal	"Harmony" agentic platform; voice, text, email for scheduling, intake, payments, care-gap closure [57]	Not public	Reach (1,000+ organizations; 2B+ interactions/yr); FedRAMP High, HITRUST; "#1 KLAS Patient Communications 2026" (vendor-reported) [57]	Reported agent task completion of 65% (vendor-reported) [57] leaves a large human workload; built for communications, not clinical navigation	Repositioned as AI agent platform (2025–26) [57]
Luma Health	Access / engagement platform + agents	Health systems, groups (multi-EHR)	Patient Success Platform; Spark AI; "Navigator" conversational agent; outreach and booking agents [58]	Not public	Broad EHR integrations (Epic, Oracle Health, MEDITECH, eClinicalWorks, athenahealth, NextGen, Greenway); UAMS automated 95% of after-hours calls (vendor-reported) [58]	Access and operations focus; clinical post-discharge navigation depth not verified	Acquired Tonic (Nov 2025) for intake/PROs and Oracle Health depth [59]; Summer '26 agents release [58]
Get Well (SAI Group)	Care-journey engagement	Health systems, VA, plans	GetWell Loop (post-discharge and episode engagement), inpatient SmartRoom, Opal on-demand patient assistant [60]	Not public	Long post-discharge track record; large customers (VA, Kaiser Permanente) [60]	Change of ownership may affect roadmap; independent outcome evidence not verified	Acquired by SAI Group (2024, per company) [60]
Fabric	Digital front door / virtual care	Health systems, plans, employers	Conversational AI plus physician-built clinical logic; symptom checking, routing, 50-state virtual clinician network [67]	Not public	Can hand off to real clinicians (own network); roll-up of Zipnosis/Gyant capabilities	Integration of five acquisitions in three years is a delivery risk (inference) [67]	Acquired UCM Digital Health (Nov 2025); Rush Connect launch (Jul 2025) [67]
Clearstep	Virtual triage / access	Health systems, federal	AI chat and voice triage on licensed Schmitt-Thompson nurse protocols; routing and scheduling [66]	Not public	Protocol-based triage (claims 97%+ clinical accuracy, vendor-reported); DHA Military Health System digital front door for 9.5M beneficiaries [66]	Built around the moment of a symptom; less focus on longitudinal adherence or transitions	DHA selection (date not verified) [66]
K Health	AI-native primary care infrastructure	Health systems	PatientGPT AI intake, 24/7 virtual primary care, EHR-integrated [69]	Not public	Partnerships with Mayo Clinic, Mass General Brigham, Cedars-Sinai, Northwell (vendor-listed) [69]	Its model needs its own virtual clinicians, which competes with the system's own staff (inference)	Repositioned as enterprise clinical AI infrastructure [69]
HealthEdge Wellframe	Payer member engagement	Health plans	Digital member engagement tied to GuidingCare care management [61]	Not public	Installed in payer care-management workflows [61]	Payer-only; generative-AI navigator capabilities not verified	HealthEdge/UST HealthProof combination announced (per company) [61]
Amwell Automated Care Programs (ex-Conversa)	Digital care programs	Health systems, plans	Automated post-discharge and post-ED programs, monitoring and alerts [63]	Not public	Established rules-based program library	Parent company has reported strategic and performance difficulties [64]; product investment uncertain	Conversa acquired with SilverCloud for a combined $320M (2021) [64]
care.coach	Avatar companion + human advocates	MA, Medicaid, PACE, VA, families	Tablet avatar with daily human check-ins, medication and appointment reminders [65]	Public: from $199/month [65]	Human in the loop; bilingual; VA-validated (vendor-reported) [65]	Needs a device; high per-member cost; outcome claims (e.g., 79% fewer ER visits) are vendor-reported [65]	—

Company	Category	Target Segment	Core Offering for this Problem	Pricing Model (if public)	Strengths	Weaknesses / Complaints	Recent Moves (2024–2026)
Lirio	Behavioural nudging	Health systems, payers, pharmacy	Behavioural-science AI "precision nudging" for adherence, screening, re-activation [62]	Not public	400M+ nudges/yr (vendor-reported) [62]	Outbound messaging, not two-way clinical navigation	Microsoft Marketplace availability [62]
Infermedica	Triage engine / API	Payers, telehealth, providers	Conversational triage, voice agent, intake; API [68]	Not public	CE Class IIb under EU MDR; customers include Teladoc (vendor-listed) [68]	EU regulatory status does not carry over to US FDA; mainly an engine for other products	Voice agent launch (date not verified) [68]
Notable	Workflow automation agents	Health systems	AI agents for intake, pre-procedure instructions, care-gap outreach [70]	Not public	Automation scale (MUSC 1.5M tasks/day, vendor-reported) [70]	Administrative focus; not a clinical navigator	CMO appointment [70]

Babylon Health is a warning for this category. It listed via SPAC at a $4.2B valuation in October 2021 and built a US value-based-care business around an AI symptom checker. Its US entity filed for Chapter 7 in August 2023, and its UK business went into administration in September 2023 [76]. Taking on risk-bearing care economics on top of AI triage, before the AI had proven value, was fatal.

User-review evidence (G2, Capterra, Gartner Peer Insights) could not be retrieved for this category in this research pass. Several weaknesses above are therefore labelled as inference and are listed for validation in 5.4.

4.3 Capability Comparison Matrix

Capability	Epic (Emmie)	Hippocratic AI	Memora (Commure)	Artera	Luma Health	Get Well	Fabric	Clearstep
Answers grounded in the patient's own record	● [54]	◐	◐	◐	◐	◐	?	?
Proactive post-discharge outreach	◐ [53]	● [51]	● [56]	◐	◐	● [60]	◐	○
Voice agent	?	● [51]	?	● [57]	● [58]	?	◐	● [66]
SMS / chat	● [53]	?	●	● [57]	● [58]	● [60]	● [67]	● [66]
Protocol-based symptom triage	?	○ (states no diagnosis) [51]	◐	?	?	?	● [67]	● [66]
Deterministic escalation to licensed clinicians	?	◐ [51]	◐	?	?	?	● (own clinicians) [67]	● [66]
Medication adherence / refill-gap support	?	◐	◐ [56]	◐	?	?	?	○
Multi-EHR deployment	○ (Epic only)	?	?	◐	● [58]	?	?	?
TCM / APCM / CCM documentation support	?	?	◐ (CCM program) [56]	?	?	?	?	?
Payer deployment	?	● [51]	?	?	○	● [60]	● [67]	◐ [66]
Independent peer-reviewed outcome evidence	?	?	?	?	?	?	?	?

● full / ◐ partial / ○ none / ? unknown (not verifiable from public sources in this research pass). The final row is "?" throughout. This research found no independent peer-reviewed trial of any of these specific products showing reduced readmissions or better adherence, but it could not rule one out.

4.4 Market Consolidation and Funding Signals

- **Consolidation into platforms.** Commure acquired Memora Health (December 20, 2024) [55]. SAI Group acquired Get Well (2024) [60]. Luma Health acquired Tonic (November 2025) [59]. Fabric made its fifth acquisition in three years with UCM Digital Health (November 2025) [67]. Hippocratic AI acquired Grove AI (January 2026) [52]. Amwell earlier combined Conversa and SilverCloud (2021, $320M combined) [64]. Standalone point navigators are being absorbed into broader engagement and operations platforms.
- **Capital is concentrating in AI-native leaders.** Hippocratic AI raised $267M across two 2025 rounds, reaching a $3.5B valuation [52]. Commure reports a $7B valuation (vendor-reported, date not verified) [55].
- **Public-market exit for chronic-care digital health.** Omada Health filed its IPO prospectus on June 9, 2025 [72]. That confirms public-market appetite for chronic-condition programs that can show outcomes.
- **Policy-driven entrants.** More than 150 digital health companies were approved for CMS ACCESS in its first cohort [26], and more than 160 organizations were participating by the time of writing [24]. Expect new outcome-paid competitors in cardiometabolic navigation, and in heart failure and COPD from 2027 [24].
- **Warning signs.** Babylon collapsed in 2023 [76]. Amwell's difficulties [64] and Epic's native patient AI [54] both put pressure on standalone vendors in Epic-heavy accounts.

5. Gap Analysis and Differentiation

5.1 Unmet Needs

Unmet Need	Evidence	Who Is Underserved	Confidence (H/M/L)
Navigation that produces billable, auditable care-management work (TCM 2-day contact, APCM access elements, RPM 2–15-day episodes) and not just conversations	TCM requires contact by the practice or its clinical staff within 2 business days [19]; APCM requires 24/7 access and digital communication [21]; no competitor publicly shows TCM/APCM documentation support (matrix row)	Primary care groups, ACOs/MSOs, FQHCs, community hospitals	M
Safety you can show: deterministic acuity escalation plus a published evaluation method	Acuity errors dominate LLM harm [38]; human–LLM interaction failures are not predicted by benchmarks [39]; ECRI #1 hazard [37]; governance committees require evidence	All provider buyers; especially risk-averse community systems	H
Record-grounded navigation for organizations not on Epic	Epic's Emmie serves Epic customers only [53][54]; 59% of individuals have multiple portals, 7% aggregate [18]	Oracle Health, MEDITECH, athenahealth and eClinicalWorks sites; multi-EHR ACOs	M
Independent outcome evidence (readmissions, adherence)	No peer-reviewed product-level trials found (matrix); transition interventions work, but recent effects are smaller [42]	All buyers, who must justify spend to CFOs and boards	M
Health-literacy-adapted, multilingual explanations of *the patient's own* regimen	28% at or below Level 1 literacy [17]; generic materials persist	Low-literacy, older and limited-English populations; Medicaid	M
State-law-configurable disclosure and consent (AB 3030, TRAIGA) plus Section 1557 bias documentation	[29][30][33] with federal pre-emption uncertainty [36]	Multi-state systems and plans	M
Payer navigation that joins pharmacy-fill adherence signals to member conversations	Part D adherence Stars measures [27]; payer segment growing fastest [48]	Regional MA and Medicaid plans without a large digital budget	L–M

5.2 Is the Gap Real? (Verdict)

Verdict: the market is crowded. The gap is in segment and execution, not a new product category.

Claimed gaps that **are not real**: - *"Nobody offers conversational AI to patients between visits."* False. Hippocratic AI, Memora/Commure, Get Well, Artera, Luma, Fabric, Clearstep, K Health and Epic all do some version of it [51]–[60][66]–[69]. - *"Nobody personalizes from the patient's record."* False for Epic customers. Ask Emmie answers questions in medical-record context [54], and care-program vendors use EHR data [56]. - *"Nobody has safety guardrails."* Leading vendors state explicit scope limits and clinician-validated testing (vendor-reported) [51][66]. Guardrails are table stakes, not a differentiator. - *"A great demo wins."* Buyers now expect EHR integration, SOC 2/HITRUST, governance evidence and ROI. A chat demo is necessary but nowhere near enough.

Gaps that **are real but partial**: 1. **Segment:** mid-size and community hospitals and primary care, ACO and FQHC organizations, especially **non-Epic sites**. They carry HRRP and APCM/TCM obligations, can't afford enterprise AI platforms, and aren't served by Epic's native AI [53][54]. 2. **Capability:** billing-aligned navigation. Workflows would record TCM contacts, APCM access and communication elements, and short RPM episodes as structured, auditable events [19][21][23]. No competitor publicly shows this. 3. **Evidence and governance:** a transparent, reusable safety evaluation (acuity test suites, reading-level checks, bias review under §92.210 [33]) and a pre-registered outcome evaluation. The whole field lacks

independent evidence. 4. **Price point:** per-member-per-month pricing that fits inside APCM/CCM fee economics (e.g., within a G0557 fee of $48.84 [21]). This is a positioning inference and needs validation.

The gap is **not** in large Epic academic systems. There, Epic plus one well-funded AI-native vendor is the likely end state.

5.3 Differentiation Opportunity for KANINI

Where KANINI could credibly win: - **Wedge: "Transitions-of-care navigator" for HRRP conditions (HF, COPD, AMI, pneumonia) in mid-size, non-Epic or Epic Community Connect hospitals and their affiliated primary care groups.** It would be sold as a governed, configurable solution that KANINI builds and integrates (not a black-box SaaS). The pitch is "measurable TCM completion and readmission tracking in 90 days." - **Position as the builder of the organization's own navigator.** Many systems want to own the patient relationship and data, and are wary of per-interaction AI vendors. KANINI's engineering-services model fits buyers who choose to build on their cloud LLM contract and EHR APIs, and who need safety guardrails, an evaluation harness and a care-team console. - **Ship compliance and evidence packs:** AB 3030/TRAIGA disclosure templates [29][30], a §92.210 bias-review worksheet [33], FDA CDS/General Wellness scoping memo [31][32], a red-team acuity test suite, and a pre-registered readmission evaluation design. These address the ECRI-level concerns [37] that slow buying. - **Payer adjacency:** extend the same engine to MA plans for Part D adherence (PDC) outreach [27], using claims and fill data rather than EHR data. Partner rather than compete with incumbent payer platforms [61]. - **ACCESS-model enablement:** provide the technology layer for ACCESS participants and co-managing primary care physicians [24], especially for the heart failure and COPD tracks starting in 2027.

Barriers to entry: - **EHR integration and marketplace access.** Epic, Oracle Health and MEDITECH write-back (scheduling, messaging, documentation) takes time and certification. Epic accounts may decide Emmie is good enough [54]. - **Security and trust credentials.** SOC 2 Type II and HITRUST are expected (competitors advertise both) [52][57]. - **Clinical content licensing.** Protocol-based triage (e.g., Schmitt-Thompson) is licensed content [66]. Staying on the navigation-and-escalation side avoids this, but it limits the scope of symptom handling. - **Regulatory scope.** Anything that looks like patient-facing triage or diagnosis may be device software under FDA policy [31]. Scope has to stay on education, navigation and escalation. - **Liability and clinical governance.** Health systems will want indemnities, incident-response processes and clinician oversight staffing, which the buyer has to fund. - **Competition for capital and attention.** Well-funded AI-native vendors [52] and 160+ ACCESS participants [24] compete for the same buyers.

5.4 Risks and Open Questions

Risk / Open Question	Why It Matters	How to Validate
Epic makes Ask Emmie widely available and adds proactive post-discharge navigation	Removes the need for a third party in Epic accounts [54]	Track Epic release notes and UGM 2026 announcements; interview 5 Epic Community Connect affiliates
A patient-harm event (missed escalation) in a KANINI deployment	Reputational and legal exposure; the field's #1 hazard [37][38]	Deterministic escalation, 100% red-team recall gate, clinician review of escalations, incident-response runbook before go-live
Readmission benefit smaller than hoped (recent interventions less effective) [42]	ROI case weakens	Pre-registered matched-cohort evaluation in the first pilot; use TCM completion and follow-up visit rate as leading indicators
AI-only contact may not satisfy the TCM "interactive contact" element, which the booklet assigns to the practitioner or clinical staff [19]	Billing claims could fail audit	Design the navigator to *support* a staff contact rather than *be* the contact; obtain coding/compliance review; consider CMS/MAC guidance request
State-law change or federal pre-emption (EO 14365) [36]	Disclosure design may need rework	Keep disclosure rules configurable per state; quarterly legal review
FDA reclassification of symptom-routing features as device CDS [31]	Regulatory burden	Formal intended-use statement; no diagnostic outputs; FDA regulatory-counsel memo

Risk / Open Question	Why It Matters	How to Validate
Market sizes disagree by ~64% and do not isolate AI navigation [47][48]	Over- or under-investment	Bottom-up buyer interviews (15–20 CFOs/VP Pop Health) on willingness to pay per enrolled patient per month
Competitor outcome claims are vendor-reported (e.g., 30% readmission reduction [51], 79% fewer ER visits [65])	Buyers may anchor on inflated benchmarks	Request references and methodology; build KANINI's own evaluation
Not verified: the "$150B annual cost of missed appointments" and BC05's "$150B avoidable utilization" composite	Frequently repeated; no primary source found	Treat as unverified; do not use in client material
Not verified: HRRP total penalty dollars and number of penalized hospitals for FY2025/FY2026 (only ≥1% counts verified) [9]	Sizing the hospital pain point	Pull the FY2026 HRRP supplemental file from CMS [84]
Not verified: Section 1557 §92.210 enforcement posture under current HHS leadership	Affects compliance urgency	Check OCR guidance; legal review
Not verified: user-review complaints (G2/KLAS/Gartner Peer Insights) for listed vendors; Hyro and Qventus positioning	Weakness column relies partly on inference	Obtain KLAS reports; run practitioner interviews
Illinois and other state limits on AI in behavioural health (not verified in this pass)	Behavioural-health use cases may be restricted	Exclude behavioural-health counselling from MVP scope; legal scan

Appendix A. Data Source and Setup

A.1 Primary Data Source

Synthea synthetic patient data (MITRE) is the primary demo dataset. It produces complete, longitudinal, fully synthetic patient records. There is no PHI. The records use real terminologies: SNOMED CT for conditions, RxNorm for medications, LOINC for observations, CVX for immunizations, and ICD-10/CPT on claims. Output formats are FHIR R4 bundles, CSV tables and optionally C-CDA, plus optional free-text clinical notes [79]. For this use case, Synthea supplies conditions (including heart failure, diabetes, COPD and hypertension), MedicationRequests, CarePlans, Encounters (including inpatient stays and discharges), Observations (vitals and labs), and synthetic clinical notes that can stand in for discharge narratives.

- Source code and releases: https://github.com/synthetichealth/synthea (latest master build published August 2026) [79]
- Configuration file: `src/main/resources/synthea.properties` [79]
- Pre-generated samples: https://synthea.mitre.org/downloads

A.2 Data Generation / Acquisition Process

1. Clone and build: `git clone https://github.com/synthetichealth/synthea.git` then `./run_synthea -p 500 -s 20260928 Massachusetts`. Use a fixed seed (`-s`) so demos can be reproduced.
2. Set these properties in `synthea.properties` (defaults verified against the current file; several are `false` by default) [79]: - `exporter.fhir.export = true` (default true) - `exporter.csv.export = true` (default **false**, so it must be enabled) - `exporter.clinical_note.export = true` (default **false**; required for discharge-narrative text) - `exporter.hospital.fhir.export = true` and `exporter.practitioner.fhir.export = true` (default true) - `exporter.years_of_history = 5` (default 10; shorter keeps demo bundles small) - `generate.only_alive_patients = true` for navigator demos (default false)
3. Filter the population to patients with at least one inpatient Encounter in the last 90 days and an HRRP-relevant condition (heart failure, COPD, AMI, pneumonia), plus a chronic-disease cohort (diabetes, hypertension) for adherence scenarios.
4. Map each medication to RxNorm and each condition to ICD-10 through the RxNav API [83] and Synthea's claims output, so grounding lookups work (MedlinePlus Connect accepts ICD-10-CM and RxNorm codes [80]).
5. Load bundles into a FHIR server (e.g., HAPI FHIR) to simulate EHR API access.

A.3 Use-Case-Specific Data Requirements

Need	Synthea resource / file	Volume	Preparation / Injection
Patient profile	Patient, Condition, MedicationRequest, CarePlan, AllergyIntolerance	100–500 patients; 8–10 curated showcase patients	Select diverse profiles: HF post-discharge, COPD, T2DM on metformin, anticoagulated AF, polypharmacy elder, Spanish-preferring patient (set language attribute)
Discharge context	Encounter (class = inpatient), DocumentReference / clinical notes	1 recent discharge per showcase patient	Write discharge instructions (daily-weight rule, follow-up within 7/14 days) into the note text
Upcoming appointments	Encounter (future) or synthetic Appointment resources	1–3 per patient	Synthea does not natively schedule future visits, so generate Appointment resources with a script
Adherence signals	MedicationRequest + synthetic fill history	12 months of fills per chronic patient	Add refill gaps to 20–30% of patients to trigger PDC <80% outreach
Vitals trend	Observation LOINC 29463-7 (weight), 85354-9 (BP panel), 4548-4 (HbA1c)	Daily weights for HF showcase patients	Add a post-discharge weight rise to trigger escalation rules

Need	Synthea resource / file	Volume	Preparation / Injection
Safety test set	Not in Synthea	150–300 scripted patient utterances	Write red-flag, off-scope, dose-change, self-harm and adversarial prompts with expected dispositions; label ground truth with a clinician
Literacy and language	Not in Synthea	Per-patient preference flag	Tag reading-level and language preference to test adaptation

A.4 Additional Reliable Data Sources

- **MedlinePlus Connect (NLM)** — https://medlineplus.gov/medlineplus-connect/ · Contains: consumer-level health topic, drug and lab information returned for ICD-10-CM, SNOMED, RxNorm and LOINC codes · Use: grounding condition and medication explanations in plain language · Format/Access: free web service (JSON/XML), no key required (verified live, e.g., I50.23 returns "Heart Failure") [80].
- **openFDA Drug Label API** — https://open.fda.gov/apis/drug/label/ · Contains: structured FDA drug labeling (indications, warnings, adverse reactions, patient counselling) · Use: grounding side-effect and "what if I miss a dose" answers · Format/Access: free REST/JSON; API key optional for higher rate limits [81].
- **DailyMed (NLM) web services** — https://dailymed.nlm.nih.gov/dailymed/app-support-web-services.cfm · Contains: SPL drug labels including Medication Guides · Use: patient-facing medication guides · Format/Access: free REST, SPL XML [82].
- **RxNav / RxNorm API (NLM)** — https://rxnav.nlm.nih.gov/ · Contains: RxNorm concepts and relationships · Use: normalizing Synthea medications (e.g., RxCUI 313988 furosemide 40 mg) · Format/Access: free REST [83].
- **CMS Provider Data Catalog — Unplanned Hospital Visits / HRRP** — https://data.cms.gov/provider-data/topics/hospitals/unplanned-hospital-visits · Contains: hospital-level readmission measures and HRRP excess readmission ratios · Use: benchmark targets and a "hospital pain" dashboard for prospect hospitals · Format/Access: free CSV/API [84].
- **NCHS National Hospital Ambulatory Medical Care Survey (NHAMCS)** — https://www.cdc.gov/nchs/namcs/documentation/ and https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Dataset_Documentation/NHAMCS/ · Contains: nationally representative ED visit microdata (reason for visit, triage acuity, disposition) · Use: calibrating ED-diversion scenarios and acuity distributions for the safety test set · Format/Access: free public-use files [15][85].
- **AHRQ HCUP Nationwide Readmissions Database (NRD)** — https://hcup-us.ahrq.gov/nrdoverview.jsp · Contains: all-payer national readmission microdata · Use: condition-level readmission baselines and cost assumptions · Format/Access: licensed purchase with data use agreement; HCUP statistical briefs free [6][7][86].
- **MyHealthfinder API (ODPHP)** — https://odphp.health.gov/myhealthfinder/api/v4/ · Contains: USPSTF-aligned preventive recommendations by age and sex · Use: preventive reminders in navigator conversations · Format/Access: free JSON API [87].
- **CMS TCM and APCM program references** — MLN908628 (TCM) and the APCM services page · Contains: billing elements and timelines · Use: encoding billable-event rules in the outreach engine · Format/Access: free PDF/web [19][21].

Appendix B. Glossary of Terms

Term	Definition
ACCESS Model	CMS Innovation Center's Advancing Chronic Care with Effective, Scalable Solutions model (began July 5, 2026), paying outcome-aligned payments for technology-supported chronic care.
ACO / MSO	Accountable Care Organization / Management Services Organization: provider entities that take on value-based contracts or provide administrative services to physician groups.
APCM	Advanced Primary Care Management: Medicare monthly codes G0556–G0558 (from 2025) for continuous, team-based primary care without time thresholds.
AB 3030	California law requiring disclaimers and human-contact instructions on generative-AI clinical communications to patients unless reviewed by a licensed provider.
CCM / PCM	Chronic Care Management / Principal Care Management: time-based Medicare care-management codes for patients with multiple or single complex chronic conditions.
Clinical Decision Support (CDS) Software	Software that supports clinical decisions; FDA guidance (Jan 2026) defines when it is and is not regulated as a device.
Deterministic Guardrail	A fixed, rule-based check that runs independently of the language model and overrides its output to enforce safety scope and escalation.
Escalation	Transfer of a patient conversation from the AI navigator to a human (care team, 911/988) when a defined trigger fires.
FHIR R4	HL7 Fast Healthcare Interoperability Resources, release 4: the API standard used to exchange patient data (e.g., MedicationRequest, CarePlan).
Flesch-Kincaid Grade Level	A readability score estimating the US school grade needed to understand a text.
Health Literacy	The ability to obtain, process and understand health information to make appropriate decisions.
HITRUST / SOC 2	Security and privacy attestation frameworks commonly required by US health care buyers.
HRRP	Hospital Readmissions Reduction Program: Medicare program reducing payments (up to 3%) to hospitals with excess 30-day readmissions for six conditions and procedures.
LLM	Large Language Model: an AI model that generates and interprets natural language.
Medicare Advantage (MA)	Privately administered Medicare health plans; included in HRRP readmission measures from FY2027.
No-show	A scheduled appointment the patient neither attends nor cancels in advance.
PDC	Proportion of Days Covered: pharmacy-claims adherence metric; PDC ≥80% is the usual adherence threshold in PQA Star measures.
PQA	Pharmacy Quality Alliance: steward of medication-use measures used in Medicare Part D Star Ratings.
RPM	Remote Physiologic Monitoring: Medicare-billable monitoring of patient physiologic data (e.g., CPT 99445, 99454, 99457, 99470).
Section 1557 (§92.210)	ACA nondiscrimination provision; §92.210 requires covered entities to identify and mitigate discrimination risk in patient care decision support tools.
Star Ratings	CMS quality ratings for Medicare Advantage and Part D plans that drive bonus payments and enrolment.
TCM	Transitional Care Management (CPT 99495/99496): post-discharge service requiring contact within 2 business days and a face-to-face visit within 7 or 14 days.

Term	Definition
TRAIGA	Texas Responsible AI Governance Act (HB 149, effective January 1, 2026), including disclosure duties when AI is used in health care services.
Synthea	Open-source synthetic patient generator from MITRE used here as the demo data basis.
RxNorm / LOINC / ICD-10-CM	Standard US terminologies for medications, laboratory/clinical observations, and diagnoses respectively.

Appendix C. References

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